

Lab Session: Server Hardening for Ubuntu OS

Prof. Mehdi Pirahandeh

October 4, 2023

Contents

1	Introduction	2
2	Prerequisites	3
3	Setting a BIOS Password	3
3.1	Objective	3
3.2	Steps	3
3.3	Example	3
4	Setting a GRUB Password	4
4.1	Objective	4
4.2	Steps	4
4.3	Example Explanation	4
5	Denying Root Access	5
5.1	Objective	5
5.2	Steps	5
5.3	Example Explanation	5
6	Creating Sudo Users	5
6.1	Objective	5
6.2	Steps	6
6.3	Example Explanation	6

7	Configuring TCP Wrappers	6
7.1	Objective	6
7.2	Steps	7
7.3	Example Explanation	7
8	Setting up Firewall/Iptables	7
8.1	Objective	7
8.2	Steps	8
8.3	Example Explanation	8
9	Installing Antivirus Software	8
9.1	Objective	8
9.2	Steps	8
9.3	Example Explanation	9
10	Securing and Auditing Logs	9
10.1	Objective	9
10.2	Steps	9
10.3	Example Explanation	10
11	Rotating Logs	10
11.1	Objective	10
11.2	Steps	10
11.3	Example Explanation	10

1 Introduction

This lab session aims to provide hands-on experience in server hardening techniques for Ubuntu OS. The following topics will be covered:

- Setting a BIOS password
- Setting a GRUB password
- Denying root access
- Creating sudo users with limited privileges
- Configuring TCP wrappers

- Setting up Firewall/IPTABLES
- Installing antivirus software
- Securing and auditing logs
- Rotating logs

2 Prerequisites

- Ubuntu Server (latest version)
- Root or sudo access

3 Setting a BIOS Password

3.1 Objective

To prevent unauthorized users from altering BIOS settings.

The BIOS is the first software that runs when a computer starts. It initializes hardware and allows you to alter system settings. Unauthorized access can lead to a compromised system.

3.2 Steps

1. Reboot the system.
2. Press the BIOS key (usually F2, F10, or DEL) to enter the BIOS setup.
3. Navigate to the Security tab.
4. Set a strong BIOS password.
5. Save and exit.

3.3 Example

No code example is available for BIOS settings as they are hardware-level configurations.

4 Setting a GRUB Password

4.1 Objective

To prevent unauthorized users from altering the GRUB bootloader.

The GRUB bootloader allows users to select which operating system or kernel to boot into. Unauthorized changes can lead to system vulnerabilities.

4.2 Steps

```
1 # Generate a GRUB password hash
2 grub-mkpasswd-pbkdf2
```

1. Open the GRUB configuration file.

```
1 sudo nano /etc/grub.d/40_custom
```

2. Add the following lines to set the password.

```
1 set superusers="root"
2 password_pbkdf2 root <your-password-hash>
```

3. Update GRUB and reboot.

```
1 sudo update-grub
2 sudo reboot
```

4.3 Example Explanation

The command ‘grub-mkpasswd-pbkdf2’ generates a hashed password, which is then added to the GRUB configuration file to secure the bootloader.

5 Denying Root Access

5.1 Objective

To minimize the probability of unauthorized intrusions by denying root access.

Root access provides complete control over the system. Denying root access minimizes the risk of unauthorized users making critical changes to the system.

5.2 Steps

```
1 # Open the SSH configuration file
2 sudo nano /etc/ssh/sshd_config
```

1. Change the following line.

```
1 PermitRootLogin no
```

2. Restart the SSH service.

```
1 sudo systemctl restart sshd
```

5.3 Example Explanation

The 'PermitRootLogin no' line in the SSH configuration file denies SSH access to the root user, thereby enhancing security.

6 Creating Sudo Users

6.1 Objective

To create sudo users and assign limited privileges.

Creating sudo users with limited privileges minimizes the risk associated with a compromised user account. It restricts the commands that can be executed, thereby limiting potential damage.

6.2 Steps

```
1 # Add a new user
2 sudo adduser <username>
```

1. Add user to sudo group.

```
1 sudo usermod -aG sudo <username>
```

2. To limit privileges, edit the sudoers file.

```
1 sudo visudo
```

3. Add or modify lines under User privilege specification.

```
1 <username> ALL=(ALL:ALL) /usr/bin/apt-get
```

6.3 Example Explanation

The ‘sudo adduser `username`’ command adds a new user. The ‘sudo usermod -aG sudo `username`’ command adds the user to the sudo group, granting them sudo privileges. The ‘sudo visudo’ command allows you to edit the sudoers file to limit the commands that the sudo user can execute.

7 Configuring TCP Wrappers

7.1 Objective

To protect the server by allowing only trusted hosts to access services like SSH, FTP, etc.

TCP wrappers provide a simple method to define which hosts are or are not allowed to connect to "wrapped" network services on your server.

7.2 Steps

```
1 # Edit the hosts.allow file
2 sudo nano /etc/hosts.allow
```

1. Add rules for SSH.

```
1      sshd : 192.168.1.
```

2. Edit the hosts.deny file.

```
1      sudo nano /etc/hosts.deny
```

3. Deny all other hosts.

```
1      sshd : ALL
```

7.3 Example Explanation

The 'sshd : 192.168.1.' line in the 'hosts.allow' file allows SSH access only from IP addresses in the 192.168.1.x range. The 'sshd : ALL' line in the 'hosts.deny' file denies SSH access from all other IP addresses.

8 Setting up Firewall/Iptables

8.1 Objective

To configure rules for incoming traffic.

Firewalls like Firewall or iptables are essential for controlling incoming and outgoing network traffic based on an organization's previously established security policies.

8.2 Steps

```
1 # Install firewalld
2 sudo apt install firewalld
```

1. Start and enable firewalld.

```
1 sudo systemctl start firewalld
2 sudo systemctl enable firewalld
```

2. Add rules.

```
1 sudo firewall-cmd --zone=public --add-port=22/tcp --
  permanent
2 sudo firewall-cmd --zone=public --add-source
  =192.168.1.0/24 --permanent
```

8.3 Example Explanation

The ‘firewall-cmd’ commands add rules to the firewall. The first command allows TCP traffic on port 22 (SSH), and the second command allows traffic only from the IP range 192.168.1.0/24.

9 Installing Antivirus Software

9.1 Objective

To install antivirus software and keep it updated.

Even Linux servers can be vulnerable to malware and viruses. Antivirus software adds an additional layer of security.

9.2 Steps


```
1 # Install ClamAV
2 sudo apt install clamav clamav-daemon
```

1. Update virus definitions.

```
1 sudo freshclam
```

9.3 Example Explanation

The ‘sudo apt install clamav clamav-daemon’ command installs ClamAV, a popular antivirus software for Linux. The ‘sudo freshclam’ command updates the virus definitions.

10 Securing and Auditing Logs

10.1 Objective

To regularly check logs for suspicious activity.

Logs contain valuable information about system activities and potential security incidents. Regular auditing is crucial for identifying and responding to security threats.

10.2 Steps

```
1 # Check auth logs
2 sudo less /var/log/auth.log
```

1. Check syslog.

```
1 sudo less /var/log/syslog
```

10.3 Example Explanation

The ‘`sudo less /var/log/auth.log`’ and ‘`sudo less /var/log/syslog`’ commands allow you to view authentication logs and system logs, respectively, which are crucial for auditing and identifying suspicious activities.

11 Rotating Logs

11.1 Objective

To keep logs for a limited period (e.g., 7 days) to save disk space.

Log files can grow indefinitely, consuming all available disk space. Log rotation helps in managing these files, ensuring that the system continues to function correctly.

11.2 Steps

```
1 # Edit logrotate configuration
2 sudo nano /etc/logrotate.conf
```

1. Add or modify the following line.

```
1 rotate 7
```

11.3 Example Explanation

The ‘`rotate 7`’ line in the ‘`logrotate.conf`’ file specifies that log files should be kept for 7 days, after which they will be rotated, thereby freeing up disk space.